

Sine rule



- 1 a False b False c True d True
- 2 No, because the sine rule requires one angle to find another angle.
- 3 Yes, because we can find the opposite angle to c using the angle sum of a triangle. Then we can find the other two sides using sine rule.
- 4 Yes, because we can find the third angle using the angle sum of a triangle. Then we can find the other two sides using sine rule.
- 5 No, because we do not know the side opposite $\angle C$, so we do not have an angle-side pair to use to find any other angles.
- 6 a $\sin A = \frac{x}{b}$ b $x = a \sin B$ c $\frac{\sin A}{a} = \frac{\sin B}{b}$
-

Unknown sides

- 7 $h = 86.52$
- 8 a $a = 10.5$ b $a = 6.9$ c $x = 13.1$ d $x = 12.9$
e $y = 8.7$ f $y = 4.4$
- 9 a $a = 16.66$ b $a = 16.66$
- 10 a 7.2 mm b 19.4 mm
- 11 a $HK = 733.17 \text{ km}$ b $KJ = 444.55 \text{ km}$



Unknown angles

12

a $x = 30.4^\circ$

b $x = 32.7^\circ$

c $x = 26.0^\circ$

d $x = 43.2^\circ$

13 $14^\circ 47'$

14

a $A = 75.78^\circ$

b $b = 3.152\text{m}$

c $c = 5.723\text{m}$

15

a $\angle OBA = 39^\circ$

b $k = 21.76$

16

a $x = 140.42^\circ$

b $x = 133.13^\circ$

c $x = 149.12^\circ$

17 $\theta = 140^\circ 40'$

Ambiguous case

18

a $8 < \text{Length} < 10$

b $\text{Length} = 8, \text{Length} \geq 10$

c $\text{Length} < 8$

19

a $\theta = 26^\circ$

b 8.7385

c 9.1247

d No, according to sine rule, $\frac{8.4}{\sin 106^\circ} = \frac{4.0}{\sin 26^\circ}$. Since they are not equal, this triangle is not possible.

20

a No

b No

c Yes

d No

21

a 0

b 1

c 1

d 2

22

a No

b No

c Yes

d No

e Yes

f Yes

g No

h Yes

i Yes



23

a $x = 41$

b 139

24

a

i It does exist

ii It could be either acute or obtuse.

b

i Does not exist.

ii No solution

25

a $x = 72.53^\circ$

b $x = 107.47^\circ$

26 $3.5 < BC < 5$

Applications

27

a $\angle WST = 136^\circ$

b $SW = 234$ km

c $h = 102.58$ km

28

a $\angle ABC = 21.89^\circ$

b $\angle BCA = 123.11^\circ$

c $AB = 29.2$ km

29

a $\angle A = 38^\circ$

b $c = 22.71$ m

c $b = 22.71$ m

d 1.9 min

30

a $a = 74^\circ$

b $d = 28.35$ m

c $h = 16.66$ m

31

a $\angle ADB = 23^\circ$

b $a = 35.65$ m

c $h = 20.4$ m

32

a $\angle ACB = 102^\circ$

b $h = 226.55$ m

c $h = 117$ m

33



a $BC = 307.3 \text{ m}$

b $AC = 335.5 \text{ m}$

c 41 m/min